WORKSHEET

Chapter 8: Boxplots 2

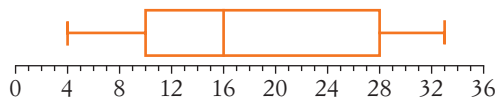
1 For each of the following boxplots list the five number summary of the data represented:

a



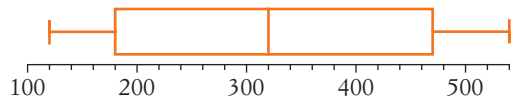
Min
Q_1
Med
Q_3
Max

b



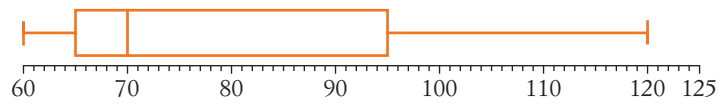
Min
Q_1
Med
Q_3
Max

c



Min
Q_1
Med
Q_3
Max

d



Min
Q_1
Med
Q_3
Max

2 Sachin's batting scores over the cricket season were as follows:

2 16 7 0 21 18 6 10 0 9 11 0 0 7 23 15 17 35 24 41 15 33 19

a List the scores in ascending order.

b Find the five number summary for Sachin's batting scores.

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c Represent this data in a boxplot.

3 A primary school teacher recorded how many students in her class brought fruit to school each day for 3 weeks. The results are below:

11 18 21 22 21 25 23 25 22 19 20 22 21 24 22

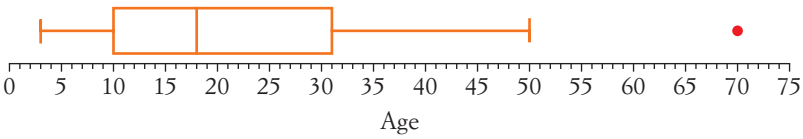
a Find the five number summary for this data.

b Calculate the interquartile range.

c Use the formula to identify any outliers.

d Represent this data as a boxplot.

4 The following boxplot shows the ages of patients admitted to the Emergency Department at a local hospital for sport-related injuries. Use it to answer the questions below:



a List the five number summary of the data.

b What is the interquartile range?

c Use the formula to confirm that there is an outlier at 70.

5 The data recorded below shows the number of laps swum in a week by members of a swimming club.

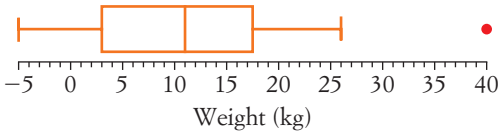
40 175 50 100 210 28 0 67 221 111 89 115 71 182 190 29 97 40 128 133 59 116

a Find the five number summary of this data.

b Use the formula to determine if there are any outliers.

c Represent this data in a boxplot.

6 Being overweight is a major risk factor for Type 2 Diabetes. Suri, a nursing student, conducted a survey among 50 adults with Type 2 diabetes, recording how many kilograms heavier they were than their ideal weight. Suri drew the boxplot below showing the data she collected.



a List the five number summary for Suri's data.

b Use the formula to determine whether Suri was correct in her placement of the outlier at 40.

7 If $Q_1 = 8.79$ and $Q_3 = 8.98$, determine which of the following values are outliers:

8.82, 9.14, 9.19, 9.26, 9.05, 8.60, 8.81, 8.72, 8.99, 9.30, 8.58

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8 The table below lists the length (in centimetres) of two types of fish caught during a fishing competition.

Bream	Flathead
33	41
37	36
25	82
29	46
26	40
31	52
25	39
39	38
28	57
26	42
39	68
35	36
31	50
27	45
40	66
33	42
35	51
36	39
42	63
25	55
29	38
33	44
27	
31	

a For both types of fish

i determine the five number summary.

	Bream	Flathead
Min		
Q_1		
Median		
Q_3		
Max		

ii use the formula to determine if there are any outliers.

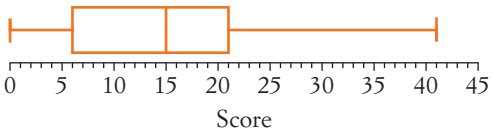
b On the same set of axes, draw a boxplot of each data set.

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Answers

1 **a** 0, 3, 5, 6, 7
b 4, 10, 16, 28, 33
c 120, 180, 320, 470, 540
d 60, 65, 70, 95, 120

2 **a** 0, 0, 0, 0, 2, 6, 7, 7, 9, 10, 11, 15, 15, 16, 17, 18, 19, 21, 23, 24, 33, 35, 41
b 0, 6, 15, 21, 41
c

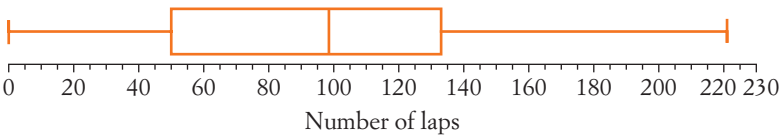


3 **a** 11, 20, 22, 23, 25
b 3
c $15.5 \leq x \leq 27.5$, 11.5 is an outlier as it is smaller than 15.5
d



4 **a** 3, 10, 18, 33, 65
b 23
c $Q_3 + 1.5 \times \text{IQR} = 67.5 \therefore 70$ is an outlier.

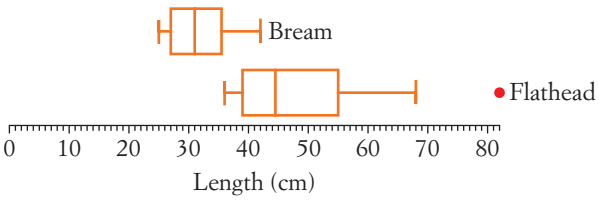
5 **a** 0, 50, 98.5, 133, 221
b $-74.5 \leq x \leq 257.5$, no outliers as all values lie within the range
c



6 **a** −5, 3, 11, 17.5, 40
b $17.5 + 1.5 \times 14.5 = 39.25$
$\therefore$ the outlier of 40 is correct

7 9.30

8 **a** **i** Bream: 25, 27, 31, 35.5, 42
Flathead: 36, 39, 44.5, 55, 82
ii Bream: $14.25 \leq x \leq 48.25$, therefore no outliers
Flathead: $15 \leq x \leq 79$, therefore 82 is an outlier
b



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